Sajid Mostafiz Noor

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Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering (CSE) 2022 – 2026
CGPA: 3.70 / 4.00

Publications

Forecasting Occupational Survivability of Rickshaw Pullers Under a Changing Climate 2023 – 2025
Rahaman, M., Hasana, M., Rahman, S. S., ***Noor, M. S. M.****, Abedin, R. R., Tahmid, M. T., Parris, D. W., Choudhury, T., Al Islam, A. B. M., & Rahman, T*
ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (**doi**: 10.1145/3770712)

- Analyzes wearable sensor data to characterize physiological heat stress among rickshaw pullers under extreme heat conditions.
- Develops Bayesian regression models and integrates CMIP6 climate projections to forecast occupational heat risk and long-term climate impacts on survivability.
- Combines statistical modeling, result validation, and thematic analysis of structured interviews to connect physiological measurements with rickshaw pullers' lived experiences.

Real Time Machine Learning Based Detection of Aggressive Driving using Feature Optimization 2025 – 2026
Ibnul Islam, ***MD Sajid Mostafiz Noor****, Ashikur Rahman, Raqeebir Rab, Abderrahmane Leshob*
International Conference on Intelligent Multimedia, Networking, and Security (**doi**: 10.1109/IMNS67862.2026.11655308)

- Develops a real-time machine learning system for aggressive driving detection using optimized sensor features.
- Compares three feature selection approaches - Feature Inclusion, Feature Exclusion, and Feature Inclusion with Group Sequencing to reduce input dimensionality.
- Achieves 98.54% accuracy with 66% fewer features and introduces a feature-sparsity-aware inference mechanism to maintain reliable predictions when sensor data are missing.

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka 2025–2026
ACM International Symposium on Wearable Computers (ISWC) (**doi**: 10.1145/3830727.3834827)

- Designs simple, threshold-based visualizations to communicate physiological heat-stress risks derived from wearable data to low-literacy informal workers.
- Introduces a mixed-method study with 11 rickshaw pullers in Dhaka, using pre- and post-assessments to capture how their perceived heat risk shifted after the visualizations.
- Analyzes interview data to identify how workers interpret visualized heat-stress information and suggests design insights for just-in-time and post-hoc heat-safety interventions.

From Wearable Data to Risk Literacy: Visualizing Heat Stress among Rickshaw Pullers through Research-Mediated Bar Charts and Potentially Self-Explanatory Rickshaw Arts
Under Review at ACM CHI 2027

- Extends prior UbiComp/ISWC research through a case study exploring how physiological heat-stress data can be translated into culturally familiar rickshaw art for Dhaka's rickshaw pullers.
- Compares four approaches to creating data-informed rickshaw art, combining artists, researchers, and AI to examine how different design processes communicate health risks.
- Evaluates how rickshaw pullers interpret the artwork, finding that familiar visual elements support understanding of heat-stress risks, while abstract data representations often require further explanation.

Research Experience

Next-Generation Computing Lab 2024 – Present
Department of Computer Science and Engineering, BUET

- Wearable sensor and environmental data collection, and the conduct of surveys and thematic analysis for ongoing interdisciplinary research projects.
- Visualization and analysis of wearable sensor data to study the effects of extreme heat on rickshaw pullers and interpret physiological stress patterns.

Work Experience

Lecturer, Department of Information and Communication Engineering (ICE) August 2026 – Present
Daffodil International University (DIU) Dhaka, Bangladesh

Projects

Bangla Biology Textbook Retrieval Benchmark for Educational QA (GitHub) January 2026 – May 2026

- Built a Bangla textbook retrieval benchmark for educational question answering using 3 HSC Biology chapters with 166 textbook nodes and 1,320 question–answer pairs.
- Trained and evaluated BM25, Multilingual-E5, and hybrid retrieval models using PyTorch and Hugging Face Transformers.
- Compared different chunking methods, hard-negative mining, and fusion techniques through controlled experiments, achieving the best result with hybrid MinMax fusion, reaching 0.746 MRR and 0.665 Recall@1.

BUET Fantasy Premier League (Live) July 2025 – September 2025

- Built a fantasy sports platform modeled after the Premier League FPL system for managing fantasy teams based on real players.
- Implemented backend services, fantasy point calculation algorithms, and leaderboard synchronization using RESTful APIs.
- Enabled users to manage teams, make team changes, make transfers, view player statistics, fixtures, and match results.
- Deployed the platform on Vercel using Github actions for CI/CD with optimized performance for an improved user experience.
- Supports a growing user base of 100+ registered users, including 82 active users involved in fantasy management.

ExamCraft – AI-driven Exam Preparation Platform (GitHub) February 2025 – August 2025

- Developed an AI-powered platform for quizzes, flashcards, and mock exams for personalized learning.
- Implemented AI-based question generation and spaced-repetition for adaptive study planning.
- Added spaced-repetition learning and performance analytics for custom recommendations and progress tracking.
- Automated build and deployment processes with GitHub Actions and deployed the containerized application on a virtual machine.

C++ Compiler (GitHub) January 2024 - March 2024

- Built a compiler from scratch in C++ using Flex (lexical analysis) and Bison (parsing).
- Designed grammar rules, symbol table management, and semantic analysis with proper error detection and reporting.
- Generated intermediate three-address code as a machine-independent representation for execution and optimization.

Grades Information System (GitHub) June 2023 – September 2023

- Designed a role-based academic management system with modular student, teacher, and admin services ensuring controlled data access.
- Engineered core backend logic for course enrollment with prerequisite validation, grade management, and result processing.
- Created an admin dashboard with CRUD operations for managing students, teachers, courses, notices, and exam schedules, and implemented database logging to ensure data consistency and track system changes.

Awards & Scholarships

- **Dean's List Scholarship**, BUET – January 2022 (Level-1, Term-1) ; July 2022 (Level-1, Term-2)
- **Board Scholarships**, Dhaka Education Board - Secondary School Certificate (SSC) and Higher Secondary Certificate (HSC)

Skills

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript
AI & Machine Learning: Machine Learning, Deep Learning, TensorFlow, PyTorch, Scikit-learn
Frontend, Backend & APIs: React, Next.js, Tailwind CSS, Node.js, NestJS, FastAPI, REST APIs
Databases & DevOps: MySQL, MongoDB, Docker, Git, GitHub Actions